

# 41107 Marine and Ocean Engineering – Formula Sheet

Condensed from the uploaded lecture notes, exam scripts, and practice material.

## GENERAL / CONSTANTS

|                     |                                            |
|---------------------|--------------------------------------------|
| Gravity             | $g = 9.81 \text{ m/s}^2$                   |
| Water density       | $\rho \approx 1025 \text{ kg/m}^3$         |
| Kinematic viscosity | $\nu \approx 10^{-6} \text{ m}^2/\text{s}$ |
| Circle area         | $A = \pi(D/2)^2 = \pi D^2/4$               |
| Angular frequency   | $\omega = 2\pi/T = 2\pi f$                 |
| Frequency           | $f = \omega/(2\pi) = 1/T$                  |
| Period              | $T = 1/f = 2\pi/\omega$                    |
| Knots to m/s        | $U[\text{m/s}] = 0.51444 U[\text{knots}]$  |
| Wave celerity       | $c = L/T = \omega/k$                       |
| Wave number         | $k = 2\pi/L$                               |

## FLOW AROUND CYLINDERS: STEADY CURRENT

|                      |                                                        |
|----------------------|--------------------------------------------------------|
| Reynolds number      | $Re = U_c D / \nu$                                     |
| Dynamic head         | $q = \frac{1}{2} \rho U^2$                             |
| Pressure coefficient | $C_p = \frac{p - p_0}{\frac{1}{2} \rho U^2}$           |
| Stagnation pressure  | $p_s - p_0 = \frac{1}{2} \rho U^2$                     |
| Mean drag            | $F_D = \frac{1}{2} \rho C_D D U^2$                     |
| Lift                 | $F_L = \frac{1}{2} \rho C_L D U^2$                     |
| Drag coefficient     | $C_D = \frac{F_D}{\frac{1}{2} \rho U^2 D}$             |
| Osc. drag coeff.     | $C'_D = \frac{F'_D}{\frac{1}{2} \rho U^2 D}$           |
| Osc. lift coeff.     | $C'_L = \frac{F'_L}{\frac{1}{2} \rho U^2 D}$           |
| Strouhal number      | $St = \frac{f_v D}{U}$                                 |
| Drag-force frequency | $f_{D,osc} = 2f_v$                                     |
| Drag dominance       | For $Re \gtrsim 10^4$ , form drag $\gg$ friction drag. |

## FLOW AROUND CYLINDERS: OSCILLATORY FLOW

|                         |                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------|
| Plane oscillatory flow  | $U(t) = U_m \sin(\omega t)$                                                                                    |
| Wave Reynolds number    | $Re = U_m D / \nu$                                                                                             |
| Keulegan–Carpenter      | $KC = \frac{U_m T}{D} = \frac{2\pi a}{D}$                                                                      |
| Horizontal excursion    | $U_m = 2\pi a / T$                                                                                             |
| Morison force           | $F_x = \frac{1}{2} \rho C_D D U  U  + \rho C_M A \ddot{U}$                                                     |
| General Morison         | $F_x \approx \frac{1}{2} \rho C_D D (U - U_b)  U - U_b  + \rho c_m A (\ddot{U} - \dot{U}_b) + \rho A \ddot{U}$ |
| Froude–Krylov           | $F_{FK} = \rho A \ddot{U}$                                                                                     |
| Coeff. relation         | $C_M = c_m + 1$                                                                                                |
| For cylinder, pot. flow | $c_m = 1 \Rightarrow C_M = 2$                                                                                  |
| Inertia/drag ratio      | $\frac{F_{I,max}}{F_{D,max}} = \frac{\pi^2}{KC} \frac{C_M}{C_D}$                                               |
| Rule of thumb           | Small $KC$ : inertia-dominated. Large $KC$ : drag-dominated.                                                   |
| Very small $KC$         | Flow may have no separation.                                                                                   |
| $KC > 7$                | Vortex shedding regime appears.                                                                                |

## VORTEX-INDUCED VIBRATIONS (VIV)

|                        |                                                                        |
|------------------------|------------------------------------------------------------------------|
| Basic oscillator       | $m\ddot{y} + c\dot{y} + ky = F$                                        |
| With added mass        | $(m + m')\ddot{y} + c\dot{y} + ky = F$                                 |
| Added mass             | $m' = \rho c_m A$                                                      |
| Undamped nat. freq.    | $\omega_n = \sqrt{\frac{k}{m + m'}}$                                   |
| Natural frequency      | $f_n = \omega_n / (2\pi)$                                              |
| Free decay             | $y(t) = A_y e^{-\zeta \omega_d t} \cos(\omega_d t)$                    |
| Damped nat. freq.      | $\omega_d = \omega_n \sqrt{1 - \zeta^2}$                               |
| Damping ratio          | $\zeta = \frac{2(m + m')\omega_n}{U}$                                  |
| Free-decay damping     | $\zeta \approx \frac{1}{2\pi} \ln \left( \frac{y_n}{y_{n+1}} \right)$  |
| Reduced velocity       | $V_r = \frac{U}{D f_n}$                                                |
| Mass ratio             | $M = \frac{m + m'}{\rho D^2} = \frac{m}{\rho D^2} + \frac{\pi}{4} c_m$ |
| Stability parameter    | $K_s = 2M(2\pi\zeta_s) = 4\pi M\zeta_s$                                |
| Simple amplitude model | $A/D \approx \frac{1.5}{K_s}$                                          |
| Cross-flow lock-in     | Typically around $4 \lesssim V_r \lesssim 8$ .                         |
| Frequency match        | In lock-in: $f \approx f_v \approx f_n$ .                              |
| Typical amplitudes     | Cross-flow $A_y$ usually larger than in-line $A_x$ .                   |

## LIFT FREQUENCY IN WAVES

|                               |                                                  |
|-------------------------------|--------------------------------------------------|
| Wave frequency                | $f_w = 1/T_w$                                    |
| Normalized lift freq.         | $N_L = f_L / f_w$                                |
| Normalized vibration freq.    | $N = f / f_w$                                    |
| Freq. relation                | $\frac{f_n}{f_w} = \frac{KC}{V_r}$               |
| Single-pair regime            | Typically $N_L \approx 2$ .                      |
| Double-pair regime            | Typically $N_L \approx 3$ .                      |
| Three-pair regime             | Typically $N_L \approx 4$ .                      |
| Small $KC$                    | Usually $f_L \neq f_v$ because of flow reversal. |
| Large $KC \rightarrow \infty$ | Steady-current limit: $f_L \rightarrow f_v$ .    |

## DIFFRACTION / LARGE BODIES

|                            |                                                        |
|----------------------------|--------------------------------------------------------|
| Diffraction regime         | Important when $D/L \gtrsim O(0.2)$ .                  |
| Typical region             | Large body + waves: often $0 \lesssim KC \lesssim 2$ . |
| Potential flow             | $\nabla^2 \phi = 0$                                    |
| Velocity potential         | $u = \phi_x, v = \phi_y, w = \phi_z$                   |
| No-flow through body       | $\partial \phi / \partial n = 0$                       |
| Linear pressure            | $p = -\rho \partial \phi / \partial t$                 |
| Total potential            | $\phi = \phi_i + \phi_s$                               |
| Horizontal force           | $F_x = - \int_0^{2\pi} p(r_0 d\theta) \cos \theta$     |
| Equivalent inertia form    | $F_x = \rho C_M (\pi r_0^2) \dot{u}$                   |
| Diffraction inertia coeff. | $C_M = \frac{4A(kr_0)}{\pi(kr_0)^2}$                   |
| Small-cylinder limit       | $kr_0 \rightarrow 0 \Rightarrow C_M \rightarrow 2$     |
| Froude–Krylov (small cyl.) | $F_{FK} = \rho A \dot{u}$                              |

## CORRELATION LENGTH

|                    |                                                                             |
|--------------------|-----------------------------------------------------------------------------|
| Correlation coeff. | $R(z) = \frac{p'(\zeta) p'(\zeta + z)}{\sqrt{p'^2(\zeta) p'^2(\zeta + z)}}$ |
| Correlation length | $L_c = \int_0^\infty R(z) dz$                                               |
| Interpretation     | Large $L_c$ : loads fluctuate coherently over a longer span.                |

## REGULAR WAVES: BASIC THEORY

|                          |                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Surface elevation        | $\eta(x, t) = \frac{H}{2} \cos(\omega t - kx)$                                                                           |
| Amplitude                | $\zeta_a = H/2$                                                                                                          |
| Angular frequency        | $\omega = 2\pi/T$                                                                                                        |
| Wave number              | $k = 2\pi/L$                                                                                                             |
| Celerity                 | $c = L/T = \omega/k$                                                                                                     |
| Dispersion relation      | $\omega^2 = gk \tanh(kh)$                                                                                                |
| Deep water               | $L = \frac{gT^2}{2\pi} = \frac{2\pi g}{\omega^2}$                                                                        |
| Horiz. particle vel.     | $u = \frac{\pi H}{T} \frac{\cosh(k(z+h))}{\sinh(kh)} \cos(\omega t - kx)$                                                |
| Vert. particle vel.      | $w = \frac{\pi H}{T} \frac{\sinh(kh)}{\cosh(k(z+h))} \sin(\omega t - kx)$                                                |
| Horiz. motion amplitude  | $a = \frac{H}{2} \frac{\cosh(k(z+h))}{\sinh(kh)}$                                                                        |
| Surface amplitude        | $a_{z=0} = \frac{H}{2 \tanh(kh)}$                                                                                        |
| Regular-wave energy      | $E_{pot} = \frac{1}{4} \rho g \zeta_a^2 L, E_{kin} = \frac{1}{4} \rho g \zeta_a^2 L, E = \frac{1}{2} \rho g \zeta_a^2 L$ |
| Breaking limit           | $\frac{H}{L} \lesssim 0.14 \tanh(kh)$                                                                                    |
| Max height on horiz. bed | $H_{max} \approx 0.14 \tanh(kh) L$                                                                                       |

## IRREGULAR WAVES / SPECTRA

|                        |                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Wave spectrum          | $S_\zeta(\omega_n) \Delta\omega = \frac{1}{2} a_n^2$                                                                      |
| Variance               | $\sigma^2 = \int_0^\infty S_\zeta(\omega) d\omega = m_0$                                                                  |
| Spectral moment        | $m_n = \int_0^\infty \omega^n S_\zeta(\omega) d\omega$                                                                    |
| Significant height     | $H_s = H_1/3 = 4\sqrt{m_0}$                                                                                               |
| Mean energy period     | $T_E = 2\pi m_{-1} / m_0$                                                                                                 |
| Mean period            | $T_1 = 2\pi m_0 / m_1$                                                                                                    |
| Zero-upcrossing period | $T_z = 2\pi \sqrt{m_0 / m_2}$                                                                                             |
| Crest-to-crest period  | $T_C = 2\pi \sqrt{m_2 / m_4}$                                                                                             |
| Peak period            | $T_p = 2\pi / \omega_p, \omega_p = \arg \max S(\omega)$                                                                   |
| Freq. conversion       | $S(\omega) d\omega = S(f) df, \omega = 2\pi f, S(f) = 2\pi S(\omega)$                                                     |
| Bretschneider          | $S(\omega) = \frac{A}{\omega^5} \exp\left(-\frac{B}{\omega^4}\right)$                                                     |
| Bretschneider coeffs.  | $A = \frac{H_s^2}{4\pi} \left(\frac{2\pi}{T_z}\right)^4, B = \frac{1}{\pi} \left(\frac{2\pi}{T_z}\right)^4$               |
| PM spectrum            | $S(\omega) = 8.1 \times 10^{-3} \frac{g^2}{\omega^5} \exp\left[-0.74 \left(\frac{g}{V\omega}\right)^4\right]$             |
| JONSWAP                | $S_J(\omega) = \alpha \frac{g^2}{\omega^5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^r$ |
| JONSWAP peak factor    | $r = \exp\left[-\frac{(\omega - \omega_p)^2}{2\sigma^2 \omega_p^2}\right]$                                                |

## PROBABILITY / RAYLEIGH

|                           |                                                                               |
|---------------------------|-------------------------------------------------------------------------------|
| Surface elevation         | Usually modeled as Gaussian with zero mean.                                   |
| Wave heights / crests     | Often modeled as Rayleigh (narrow-banded sea).                                |
| Peak-amplitude exceedance | $P(X > x) = \exp\left(-\frac{x^2}{2m_0}\right)$                               |
| Crest exceedance          | $P(\xi_c > h) = \exp\left(-\frac{h^2}{2m_0}\right)$                           |
| Wave-height exceedance    | $P(H > h) = \exp\left(-\frac{h^2}{8m_0}\right)$                               |
| Expected no. exceedances  | Approx. $N_{exc} = N P(\cdot)$ , with $N \approx$ no. of waves in the record. |

| SHIP MOTION DYNAMICS / REGULAR WAVES  |                                                                                                                                                                           | SHIP HYDROSTATICS / FLOATATION |                                                                                                     | STRENGTH OF SHIPS / BENDING             |                                                                                                                                                                    |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 DOF                                 | Surge, sway, heave, roll, pitch, yaw.                                                                                                                                     | Buoyancy = weight              | $\rho g \nabla = W$ , and since $W = mg$ , a floating body has $m = \rho \nabla$ .                  | Distributed load                        | $q(x) = b(x) - w(x)$ ; $b$ = buoyancy line load upward, $w$ = weight line load downward.                                                                           |
| Linear theory                         | Response proportional to wave amplitude and at same frequency, with phase shift.                                                                                          | Box / barge displacement       | $\nabla = LBT$ .                                                                                    | Total load from diagram                 | $W = \int w(x) dx$ . Area under a line-load diagram gives total force / weight.                                                                                    |
| Excitation + radiation                | $F(t) = F_{exc}(t) + F_{rad}(t) + F_{rest}(t)$ .                                                                                                                          | Draught of box/barge           | $T = W/(\rho g LB)$ .                                                                               | Triangle / rectangle loads              | $W_{tri} = 0.5 \times \text{base} \times \text{peak}$ , $W_{rect} = qL$ .                                                                                          |
| Vertical cylinder EOM                 | $(m + a)\ddot{z} + b\dot{z} + cz = F_a \cos(\omega t + \epsilon_{F\zeta})$ .                                                                                              | Weight from cargo only         | $W = \rho_s g (LBD_{cargo})$ .                                                                      | Still-water load                        | Buoyancy upward minus weight downward.                                                                                                                             |
| Waterplane area                       | $A_w = \pi D^2/4$ for a vertical circular cylinder.                                                                                                                       | Center of buoyancy             | $KB = T/2$ for rectangular / box-like sections.                                                     | Shear relation                          | $dV/dx = -q(x)$ .                                                                                                                                                  |
| Restoring coeff.                      | $c = \rho g A_w = \rho g \pi D^2/4$ .                                                                                                                                     | Waterplane inertia             | $I_x = LB^3/12$ (transverse), or in general $I_x = (2/3) \int y^3(x) dx$ .                          | Bending-moment relation                 | $dM/dx = V(x)$ .                                                                                                                                                   |
| Floating-cylinder mass                | $m = \rho \nabla$ , with draught $h$ and displaced volume $\nabla = A_w h$ .                                                                                              | Metacentric radius             | $BM_T = I_x/\nabla$ .                                                                               | Neutral axis                            | $z_n = (\sum A_i z_i)/(\sum A_i)$ .                                                                                                                                |
| Undamped nat. freq.                   | $\omega_n = \sqrt{c/(m + a)}$ ; if $a$ is neglected for a floating vertical cylinder, $\omega_n = \sqrt{g/h}$ .                                                           | Metacentric height from keel   | $KMT = KB + BM_T$ .                                                                                 | Thin-walled area                        | $A_l = l_i t_i$ ; $l_i$ = plate length, $t_i$ = plate thickness.                                                                                                   |
| Harmonic response                     | $z = z_a \cos(\omega t + \epsilon_{z\zeta})$ .                                                                                                                            | Metacentric height             | $GMT = KMT - KG$ .                                                                                  | Uniform-thickness short-cut             | If all plates have the same thickness $t$ , then $z_n = (\sum l_i z_i)/(\sum l_i)$ .                                                                               |
| Velocity                              | $\dot{z} = -z_a \omega \sin(\omega t + \epsilon_{z\zeta})$ , $\ddot{z} = -z_a \omega^2 \cos(\omega t + \epsilon_{z\zeta})$ .                                              | Righting lever (small heel)    | $GZ \approx GMT \sin \phi$ .                                                                        | Centroid level                          | $z_i$ = vertical coordinate of the centroid of element $i$ from the chosen datum. Horizontal plate: centre at its own level; vertical plate: centre at mid-height. |
| RAO magnitude                         | $z_n(\omega) =  \Phi(\omega) \zeta_a$ , with wave amplitude $\zeta_a = H/2$ .                                                                                             | Restoring moment               | $M_R = \rho g \nabla GZ$ .                                                                          | Second moment of area                   | $I = \int z^2 dA$ .                                                                                                                                                |
| Complex RAO                           | $\Phi(\omega) = \Re\Phi + i\Im\Phi$ .                                                                                                                                     | Wall-sided formula             | $GZ = \sin \phi \left( GMT + (1/2)BMT \tan^2 \phi \right)$ .                                        | Steiner / parallel-axis                 | $I \approx \sum (I_i + A_i a_i^2)$ .                                                                                                                               |
| RAO amplitude                         | $ \Phi  = \sqrt{(\Re\Phi)^2 + (\Im\Phi)^2}$ .                                                                                                                             | Wall-sided moment              | $M_R = \rho g \nabla \sin \phi \left( GMT + (1/2)BMT \tan^2 \phi \right)$ .                         | Thin-walled approx.                     | $I \approx \sum l_i t_i z_i^2$ .                                                                                                                                   |
| RAO phase                             | $\epsilon = \text{atan2}(\Im\Phi, \Re\Phi)$ .                                                                                                                             | Metacentre in WL               | If $KMT = T$ for a rectangular barge, then $T = B/\sqrt{6}$ .                                       | Navier bending stress                   | $\sigma_{11} = M_y z / I$ .                                                                                                                                        |
| Reference angle                       | $\alpha = \tan^{-1}( \Im\Phi/\Re\Phi )$ . Q1 $\theta = \alpha$ , Q2 $\theta = 180^\circ - \alpha$ , Q3 $\theta = 180^\circ + \alpha$ , Q4 $\theta = 360^\circ - \alpha$ . | Stability criterion            | Initial stability if $GMT > 0$ ; unstable if $GMT < 0$ .                                            | Section modulus                         | $Z = I/z_{max}$ .                                                                                                                                                  |
| FORWARD SPEED / ENCOUNTERED WAVES     |                                                                                                                                                                           | SHIP RESISTANCE / SCALING      |                                                                                                     | Max bending stress                      | $\sigma_{max} = M/Z$ .                                                                                                                                             |
| Encounter angle                       | $\mu$ = wave-encounter angle used in the course (typically $90^\circ$ to $270^\circ$ from beam to head seas).                                                             | Froude number                  | $Fn = U/\sqrt{gL}$ .                                                                                | Stress location                         | Largest $ \sigma $ occurs furthest from the neutral axis.                                                                                                          |
| Encounter freq.                       | $\omega_e = \omega - (\omega^2 U/g) \cos \mu$ .                                                                                                                           | Reynolds number                | $Re = UL/\nu$ .                                                                                     | Hogging / sagging                       | Load distribution sets the sign of bending moment.                                                                                                                 |
| Encounter period                      | $T_e = 2\pi/\omega_e$ .                                                                                                                                                   | ITTC friction coeff.           | $C_F = 0.075/(\log_{10} Re - 2)^2$ .                                                                | WAVE-INDUCED BENDING / SHORT-TERM STATS |                                                                                                                                                                    |
| Energy conservation                   | $S(\omega) d\omega = S^*(\omega_e) d\omega_e$ .                                                                                                                           | Friction resistance            | $R_F = (1/2)\rho U^2 SC_F$ .                                                                        | VBM response spectrum                   | $S_M(\omega_e) =  \Phi_M(\omega_e) ^2 S_e(\omega_e)$ .                                                                                                             |
| Spectrum transform                    | $S^*(\omega_e) = S(\omega)/(1 - 2\omega U \cos \mu/g)$ .                                                                                                                  | Total resistance               | $R_T = (1 + k)R_F + R_W$ .                                                                          | VBM response moments                    | $m_{R,n} = \int_{-\infty}^{\infty} \omega_e^n S_M(\omega_e) d\omega_e$ .                                                                                           |
| Useful inversion                      | Let $a = (U/g) \cos \mu$ . Then $\omega_e = \omega - a\omega^2$ and $\omega = (1 \pm \sqrt{1 - 4a\omega_e})/(2a)$ .                                                       | Effective power                | $P_E = R_T U$ .                                                                                     | Rayleigh exceedance                     | $P(X > x) = \exp[-x^2/(2m_{R,0})]$ .                                                                                                                               |
| Head seas shortcut                    | If $\mu = 180^\circ$ , then $\cos \mu = -1$ and $\omega_e = \omega + (\omega^2 U/g)$ .                                                                                    | Geometric scale                | $\alpha = L_s/L_m$ (ship / model).                                                                  | Applies to                              | Short-term prediction in a stationary sea state.                                                                                                                   |
| SPECTRAL RESPONSE OF SHIPS / FLOATERS |                                                                                                                                                                           | Area scaling                   | $S_x/S_m = \alpha^2$ .                                                                              | VERY COMMON EXAM SHORTCUTS              |                                                                                                                                                                    |
| Wave component energy                 | $S_\zeta(\omega_n)\Delta\omega = a_n^2/2$ .                                                                                                                               | Volume scaling                 | $\nabla_x/\nabla_m = \alpha^3$ .                                                                    | Stagnation pressure                     | Use $C_p = 1$ at the stagnation point, so $p - p_0 = 0.5\rho U^2$ .                                                                                                |
| Response amplitude                    | $r_0 =  \Phi(\omega) a$ .                                                                                                                                                 | Froude speed scaling           | $U_x/U_m = \sqrt{\alpha}$ .                                                                         | Oscillating drag freq.                  | Use $f_{D,osc} = 2f_v$ .                                                                                                                                           |
| Response spectrum                     | $S_R(\omega) =  \Phi(\omega) ^2 S_\zeta(\omega)$ .                                                                                                                        | Froude time scaling            | $T_x/T_m = \sqrt{\alpha}$ .                                                                         | Lift regime name                        | If $N_L = 3$ , think double-pair regime.                                                                                                                           |
| From spec. to RAO                     | At one frequency: $ \Phi  = \sqrt{S_R/S_\zeta}$ . For pitch notation often $ \Phi  = \sqrt{R/S}$ .                                                                        | Mass / weight scaling          | $W_x/W_m \approx \alpha^3$ .                                                                        | Small log / box stability               | For a floating log or box: compute $GMT = KB + BM_T - KG$ .                                                                                                        |
| Response moments                      | $m_{R,n} = \int_0^\infty \omega^n S_R(\omega) d\omega$ .                                                                                                                  | Resistance scaling             | $R_x/R_m \approx \alpha^3$ at corresponding speeds.                                                 | Response from spectrum                  | Area under response spectrum gives $m_{R,0} \Rightarrow$ significant response amplitude.                                                                           |
| Sig. response amplitude               | $r_{1/3} = 4\sqrt{m_{R,0}}$ . Area under the response spectrum gives $m_{R,0}$ .                                                                                          | Power scaling                  | $P_x/P_m \approx \alpha^{3.5}$ .                                                                    | Encountered waves                       | Use $\omega_e$ first, then convert back to $T_e = 2\pi/\omega_e$ .                                                                                                 |
| Peak exceedance                       | $P(X > x) = \exp[-x^2/(2m_{R,0})]$ .                                                                                                                                      | Model testing                  | Preserve Froude similarity in practice; exact Reynolds similarity cannot be matched simultaneously. | KC on seabed                            | Find local orbital amplitude $a$ at $z = -h$ , then $KC = 2\pi a/D$ .                                                                                              |
| Wave-height reminder                  | $H_s = 4\sqrt{m_0}$ , so $m_0 = (H_s/4)^2$ .                                                                                                                              |                                |                                                                                                     | Spectral moments                        | Useful pair: $m_0 = (H_s/4)^2$ , $T_1 = 2\pi m_0/m_1$ .                                                                                                            |
| Representative period                 | $T_1 = 2\pi m_0/m_1$ .                                                                                                                                                    |                                |                                                                                                     | Resistance scaling                      | At corresponding speeds: $\bar{R}_s/\bar{R}_m = (L_s/L_m)^3$ .                                                                                                     |
| Qualitative rule                      | Large overlap between $S_\zeta(\omega)$ and $ \Phi(\omega) $ gives large response.                                                                                        |                                |                                                                                                     | Vertical cyl. in heave                  | If floating at draught $h$ and added mass is neglected: $\omega_n = \sqrt{g/h}$ .                                                                                  |